

AI Inventory Agent - REST API Documentation

Base URL

```
http://localhost:8080/api
```

Authentication

All endpoints except `/api/auth/register`, `/api/auth/login`, and `/api/public/**` require a valid JWT token.

How to authenticate:

1. Register or login to obtain a JWT token.
2. Include the token in the `Authorization` header for all subsequent requests:

```
Authorization: Bearer <your-jwt-token>
```

Token expiry: 86,400,000 ms (24 hours).

Session policy: Stateless (no server-side sessions). Each request must carry the token.

Error Response Format

All error responses follow this structure:

```
{
  "error": "<Error Type>",
  "message": "<Human-readable message>",
  "timestamp": "2026-07-25T10:30:00"
}
```

Error types:

HTTP Status	Error Type	Description
400	Validation Failed	Request body failed validation
404	Not Found	Requested resource does not exist
500	Internal Server Error	Unexpected server error

1. Authentication (`/api/auth`)

1.1 Register a New User

```
POST /api/auth/register
```

Auth required: No (public)

Request Body:

```
{
  "username": "john_doe",
  "email": "john@example.com",
  "password": "secret123",
  "fullName": "John Doe"
}
```

Field	Type	Required	Constraints
username	String	Yes	Min 3, Max 50 characters
email	String	Yes	Must be valid email format
password	String	Yes	Min 6 characters
fullName	String	Yes	Max 100 characters

Success Response (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJqaG9uX2RvZSIsInVzZXJuYW11Ijoiamhvb19kb2UiLCJpYXQiOiJlbnV4OTcwMDAsImV4cCI6MTcyMTk4MzQwMH0",
  "username": "john_doe",
  "email": "john@example.com",
  "role": "USER",
  "fullName": "John Doe"
}
```

Error Responses:

- 400 Bad Request - Validation failed (e.g., missing fields, weak password)
- 500 Internal Server Error - Username or email already exists:

```
{
  "error": "Internal Server Error",
  "message": "Username already exists",
  "timestamp": "2026-07-25T10:30:00"
}
```

1.2 Login

POST /api/auth/login

Auth required: No (public)

Request Body:

```
{
  "username": "john_doe",
  "password": "secret123"
}
```

Field	Type	Required
username	String	Yes
password	String	Yes

Success Response (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJqaG9uX2RvZSIsInVzZXJuYW11Ijoiamhvb19kb2UiLCJpYXQiOiJlbnV4OTcwMDAsImV4cCI6MTcyMTk4MzQwMH0",
  "username": "john_doe",
  "email": "john@example.com",
  "role": "USER",
  "fullName": "John Doe"
}
```

Error Response:

```
{
  "error": "Internal Server Error",
  "message": "Invalid username or password",
  "timestamp": "2026-07-25T10:30:00"
}
```

1.3 Get Current User

```
GET /api/auth/me
```

Auth required: Yes (JWT)

Success Response (200 OK):

Returns the username of the currently authenticated user as a plain string:

```
john_doe
```

2. Products (/api/products)

All product endpoints require JWT authentication.

2.1 Get All Products

```
GET /api/products
```

Auth required: Yes

Success Response (200 OK):

```
[
  {
    "id": 1,
    "name": "Wireless Mouse",
    "description": "Ergonomic wireless mouse with USB receiver",
    "sku": "WM-001",
    "category": "Electronics",
    "unitPrice": 29.99,
    "costPrice": 15.00,
    "reorderLevel": 20,
    "imageUrl": "https://example.com/mouse.jpg",
    "isActive": true,
    "createdAt": "2026-07-01T08:00:00",
    "updatedAt": "2026-07-01T08:00:00"
  },
  {
    "id": 2,
    "name": "USB-C Cable",
    "description": "1m braided USB-C charging cable",
    "sku": "UC-002",
    "category": "Accessories",
    "unitPrice": 9.99,
    "costPrice": 3.50,
    "reorderLevel": 50,
    "imageUrl": null,
    "isActive": true,
    "createdAt": "2026-07-02T09:15:00",
    "updatedAt": "2026-07-02T09:15:00"
  }
]
```

2.2 Get Product by ID

```
GET /api/products/{id}
```

Auth required: Yes

Path Parameters:

Param	Type	Description
id	Long	Product ID

Success Response (200 OK):

```
{
  "id": 1,
  "name": "Wireless Mouse",
  "description": "Ergonomic wireless mouse with USB receiver",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 29.99,
  "costPrice": 15.00,
  "reorderLevel": 20,
  "imageUrl": "https://example.com/mouse.jpg",
  "isActive": true,
  "createdAt": "2026-07-01T08:00:00",
  "updatedAt": "2026-07-01T08:00:00"
}
```

Error Response (404):

```
{
  "error": "Not Found",
  "message": "Product not found with id: 999",
  "timestamp": "2026-07-25T10:30:00"
}
```

2.3 Create Product

POST /api/products

Auth required: Yes

Request Body:

```
{
  "name": "Wireless Mouse",
  "description": "Ergonomic wireless mouse with USB receiver",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 29.99,
  "costPrice": 15.00,
  "reorderLevel": 20,
  "imageUrl": "https://example.com/mouse.jpg"
}
```

Field	Type	Required	Constraints
name	String	Yes	Max 100 characters
description	String	No	Max 500 characters
sku	String	Yes	Unique, max 50 characters
category	String	Yes	Max 50 characters
unitPrice	Decimal	Yes	Must be positive (precision 10, scale 2)
costPrice	Decimal	Yes	Must be positive (precision 10, scale 2)
reorderLevel	Integer	Yes	Minimum 0
imageUrl	String	No	Max 255 characters

Success Response (200 OK):

```
{
  "id": 1,
  "name": "Wireless Mouse",
  "description": "Ergonomic wireless mouse with USB receiver",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 29.99,
  "costPrice": 15.00,
  "reorderLevel": 20,
  "imageUrl": "https://example.com/mouse.jpg",
  "isActive": true,
  "createdAt": "2026-07-25T10:30:00",
  "updatedAt": "2026-07-25T10:30:00"
}
```

Error Responses:

- 400 Bad Request - Validation failed or SKU already exists:

```
{
  "error": "Internal Server Error",
  "message": "Product with SKU already exists: WM-001",
  "timestamp": "2026-07-25T10:30:00"
}
```

2.4 Update Product

PUT /api/products/{id}

Auth required: Yes

Path Parameters:

Param Type Description

id Long Product ID

Request Body: (same as Create Product)

```
{
  "name": "Wireless Mouse Pro",
  "description": "Updated ergonomic wireless mouse",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 34.99,
  "costPrice": 17.00,
  "reorderLevel": 15,
  "imageUrl": "https://example.com/mouse-pro.jpg"
}
```

Success Response (200 OK):

```
{
  "id": 1,
  "name": "Wireless Mouse Pro",
  "description": "Updated ergonomic wireless mouse",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 34.99,
  "costPrice": 17.00,
  "reorderLevel": 15,
  "imageUrl": "https://example.com/mouse-pro.jpg",
  "isActive": true,
  "createdAt": "2026-07-25T10:30:00",
  "updatedAt": "2026-07-25T11:00:00"
}
```

2.5 Delete Product (Soft Delete)

```
DELETE /api/products/{id}
```

Auth required: Yes

Path Parameters:

Param Type Description

id	Long Product ID
----	-----------------

*Note: This performs a **soft delete** -- the product's `isActive` flag is set to `false`. The product is not removed from the database.*

Success Response (200 OK):

```
{
  "id": 1,
  "name": "Wireless Mouse",
  "isActive": false,
  "message": "Product deactivated successfully"
}
```

Error Response (404):

```
{
  "error": "Not Found",
  "message": "Product not found with id: 999",
  "timestamp": "2026-07-25T10:30:00"
}
```

2.6 Search Products

```
GET /api/products/search?q={query}
```

Auth required: Yes

Query Parameters:

Param Type Required

q	String Yes	Description
		Search term (name partial match)

Performs a case-insensitive partial match on product name.

Example Request:

```
GET /api/products/search?q=wireless
```

Success Response (200 OK):

```
[
  {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001",
    "category": "Electronics",
    "unitPrice": 29.99,
    "isActive": true
  },
  {
    "id": 3,
    "name": "Wireless Keyboard",
    "sku": "WK-003",
    "category": "Electronics",
    "unitPrice": 49.99,
    "isActive": true
  }
]
```

2.7 Get Low-Stock Products

```
GET /api/products/low-stock
```

Auth required: Yes

Returns all active products whose current stock level (total IN movements minus total OUT movements) is below their defined `reorderLevel`.

Success Response (200 OK):

```
[
  {
    "id": 2,
    "name": "USB-C Cable",
    "sku": "UC-002",
    "category": "Accessories",
    "unitPrice": 9.99,
    "reorderLevel": 50,
    "currentStock": 12,
    "isActive": true
  }
]
```

2.8 Get Product by SKU

```
GET /api/products/sku/{sku}
```

Auth required: Yes

Path Parameters:

Param Type	Description
sku	String Product SKU code

Example Request:

```
GET /api/products/sku/WM-001
```

Success Response (200 OK):

```
{
  "id": 1,
  "name": "Wireless Mouse",
  "description": "Ergonomic wireless mouse with USB receiver",
  "sku": "WM-001",
  "category": "Electronics",
  "unitPrice": 29.99,
  "costPrice": 15.00,
  "reorderLevel": 20,
  "imageUrl": "https://example.com/mouse.jpg",
  "isActive": true,
  "createdAt": "2026-07-01T08:00:00",
  "updatedAt": "2026-07-01T08:00:00"
}
```

Error Response (404):

```
{
  "error": "Not Found",
  "message": "Product not found with SKU: XX-999",
  "timestamp": "2026-07-25T10:30:00"
}
```

3. Inventory (/api/inventory)

All inventory endpoints require JWT authentication.

3.1 Get All Inventory Movements

GET /api/inventory/movements

Auth required: Yes

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "movementType": "IN",
    "quantity": 100,
    "referenceNumber": "PO-001",
    "notes": "Initial stock purchase",
    "createdBy": "admin",
    "movementDate": "2026-07-01T08:00:00"
  },
  {
    "id": 2,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "movementType": "OUT",
    "quantity": 5,
    "referenceNumber": "SALE-1",
    "notes": "Sale to customer: Acme Corp",
    "createdBy": "system",
    "movementDate": "2026-07-05T14:30:00"
  }
]
```

3.2 Create Inventory Movement

POST /api/inventory/movements

Auth required: Yes

Request Body:

```
{
  "productId": 1,
  "movementType": "IN",
  "quantity": 50,
  "referenceNumber": "PO-002",
  "notes": "Restock from supplier",
  "createdBy": "admin"
}
```

Field	Type	Required	Constraints
productId	Long	Yes	Must reference a valid product
movementType	String	Yes	"IN" or "OUT"
quantity	Integer	Yes	Must be positive
referenceNumber	String	No	Max 50 characters
notes	String	No	Max 500 characters
createdBy	String	No	Max 50 characters

Success Response (200 OK):

```
{
  "id": 3,
  "product": {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001"
  },
  "movementType": "IN",
  "quantity": 50,
  "referenceNumber": "PO-002",
  "notes": "Restock from supplier",
  "createdBy": "admin",
  "movementDate": "2026-07-25T10:30:00"
}
```

3.3 Get Movements by Product ID

```
GET /api/inventory/movements/product/{id}
```

Auth required: Yes

Path Parameters:

Param Type Description

id Long Product ID

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "movementType": "IN",
    "quantity": 100,
    "referenceNumber": "PO-001",
    "notes": "Initial stock purchase",
    "createdBy": "admin",
    "movementDate": "2026-07-01T08:00:00"
  },
  {
    "id": 2,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "movementType": "OUT",
    "quantity": 5,
    "referenceNumber": "SALE-1",
    "notes": "Sale to customer: Acme Corp",
    "createdBy": "system",
    "movementDate": "2026-07-05T14:30:00"
  }
]
```

3.4 Get Current Stock for a Product

```
GET /api/inventory/stock/{productId}
```

Auth required: Yes

Path Parameters:

Param	Type	Description
productId	Long	Product ID

Calculates current stock as: $SUM(IN\ movements) - SUM(OUT\ movements)$.

Success Response (200 OK):

```
{
  "productId": 1,
  "productName": "Wireless Mouse",
  "sku": "WM-001",
  "totalIn": 100,
  "totalOut": 15,
  "currentStock": 85,
  "reorderLevel": 20
}
```

4. Sales (/api/sales)

All sales endpoints require JWT authentication.

4.1 Get All Sales

GET /api/sales

Auth required: Yes

Returns all sales sorted by `saleDate` descending (most recent first).

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "quantity": 5,
    "unitPrice": 29.99,
    "totalPrice": 149.95,
    "customerName": "Acme Corp",
    "saleDate": "2026-07-25T10:30:00",
    "status": "COMPLETED",
    "createdAt": "2026-07-25T10:30:00",
    "updatedAt": "2026-07-25T10:30:00"
  }
]
```

4.2 Get Sale by ID

GET /api/sales/{id}

Auth required: Yes

Path Parameters:

Param	Type	Description
-------	------	-------------

Param Type Description

id Long Sale ID

Success Response (200 OK):

```
{
  "id": 1,
  "product": {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001"
  },
  "quantity": 5,
  "unitPrice": 29.99,
  "totalPrice": 149.95,
  "customerName": "Acme Corp",
  "saleDate": "2026-07-25T10:30:00",
  "status": "COMPLETED",
  "createdAt": "2026-07-25T10:30:00",
  "updatedAt": "2026-07-25T10:30:00"
}
```

Error Response (404):

```
{
  "error": "Not Found",
  "message": "Sale not found with id: 999",
  "timestamp": "2026-07-25T10:30:00"
}
```

4.3 Create Sale

POST /api/sales

Auth required: Yes

Request Body:

```
{
  "productId": 1,
  "quantity": 5,
  "unitPrice": 29.99,
  "customerName": "Acme Corp"
}
```

Field	Type	Required	Constraints
productId	Long	Yes	Must reference a valid product
quantity	Integer	Yes	Must be positive
unitPrice	Decimal	Yes	Must be positive
customerName	String	Yes	Max 100 characters

Business Logic: The service checks that the product has sufficient stock before creating the sale. If stock is insufficient, a 500 error is returned. On success, an inventory OUT movement is automatically recorded with reference SALE-{id}.

Success Response (200 OK):

```
{
  "id": 2,
  "product": {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001"
  },
  "quantity": 5,
  "unitPrice": 29.99,
  "totalPrice": 149.95,
  "customerName": "Acme Corp",
  "saleDate": "2026-07-25T11:00:00",
  "status": "COMPLETED",
  "createdAt": "2026-07-25T11:00:00",
  "updatedAt": "2026-07-25T11:00:00"
}
```

Error Response (500):

```
{
  "error": "Internal Server Error",
  "message": "Insufficient stock for product: Wireless Mouse. Available: 3, Requested: 5",
  "timestamp": "2026-07-25T11:00:00"
}
```

4.4 Get Sales by Date Range

```
GET /api/sales/date-range?start={start}&end={end}
```

Auth required: Yes

Query Parameters:

Param	Type	Required	Format	Description
start	String	Yes	YYYY-MM-DD (ISO)	Start date
end	String	Yes	YYYY-MM-DD (ISO)	End date

Example Request:

```
GET /api/sales/date-range?start=2026-07-01&end=2026-07-31
```

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "quantity": 5,
    "unitPrice": 29.99,
    "totalPrice": 149.95,
    "customerName": "Acme Corp",
    "saleDate": "2026-07-15T10:30:00",
    "status": "COMPLETED"
  },
  {
    "id": 3,
    "product": {
      "id": 2,
      "name": "USB-C Cable",
      "sku": "UC-002"
    },
    "quantity": 20,
    "unitPrice": 9.99,
    "totalPrice": 199.80,
    "customerName": "Tech Supplies Inc",
    "saleDate": "2026-07-20T14:15:00",
    "status": "COMPLETED"
  }
]
```

4.5 Get Total Sales Statistics

```
GET /api/sales/stats/total
```

Auth required: Yes

Returns aggregate sales data: total revenue, total number of transactions, total quantity sold, and average sale amount.

Success Response (200 OK):

```
{
  "totalRevenue": 15499.50,
  "totalTransactions": 120,
  "totalQuantitySold": 850,
  "averageSaleAmount": 129.16
}
```

4.6 Get Detailed Sales Statistics

```
GET /api/sales/stats
```

Auth required: Yes

Returns comprehensive sales statistics including recent transactions.

Success Response (200 OK):

```
{
  "totalRevenue": 15499.50,
  "totalTransactions": 120,
  "totalQuantitySold": 850,
  "averageSaleAmount": 129.16,
  "recentSales": [
    {
      "id": 118,
      "product": {
        "id": 1,
        "name": "Wireless Mouse"
      },
      "quantity": 3,
      "unitPrice": 29.99,
      "totalPrice": 89.97,
      "customerName": "GadgetWorld",
      "saleDate": "2026-07-25T09:00:00",
      "status": "COMPLETED"
    },
    {
      "id": 117,
      "product": {
        "id": 3,
        "name": "Wireless Keyboard"
      },
      "quantity": 2,
      "unitPrice": 49.99,
      "totalPrice": 99.98,
      "customerName": "Office Depot",
      "saleDate": "2026-07-24T16:45:00",
      "status": "COMPLETED"
    }
  ]
}
```

5. Purchases (/api/purchases)

All purchase endpoints require JWT authentication.

5.1 Get All Purchases

GET /api/purchases

Auth required: Yes

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "supplier": {
      "id": 1,
      "name": "TechWholesale Ltd",
      "email": "orders@techwholesale.com"
    },
    "quantity": 100,
    "unitCost": 15.00,
    "totalCost": 1500.00,
    "purchaseDate": "2026-07-01T08:00:00",
    "status": "RECEIVED",
    "createdAt": "2026-07-01T08:00:00",
    "updatedAt": "2026-07-05T12:00:00"
  }
]
```

5.2 Get Purchase by ID

GET /api/purchases/{id}

Auth required: Yes

Path Parameters:

Param Type Description

id Long Purchase ID

Success Response (200 OK):

```
{
  "id": 1,
  "product": {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001"
  },
  "supplier": {
    "id": 1,
    "name": "TechWholesale Ltd",
    "contactPerson": "Jane Smith",
    "email": "orders@techwholesale.com",
    "phone": "+1-555-0100"
  },
  "quantity": 100,
  "unitCost": 15.00,
  "totalCost": 1500.00,
  "purchaseDate": "2026-07-01T08:00:00",
  "status": "RECEIVED",
  "createdAt": "2026-07-01T08:00:00",
  "updatedAt": "2026-07-05T12:00:00"
}
```

Error Response (404):

```
{
  "error": "Not Found",
  "message": "Purchase not found with id: 999",
  "timestamp": "2026-07-25T10:30:00"
}
```

5.3 Create Purchase

POST /api/purchases

Auth required: Yes

Request Body:

```
{
  "productId": 1,
  "supplierId": 1,
  "quantity": 100,
  "unitCost": 15.00,
  "purchaseDate": "2026-07-25T10:00:00"
}
```

Field	Type	Required	Constraints
productId	Long	Yes	Must reference a valid product
supplierId	Long	Yes	Must reference a valid supplier
quantity	Integer	Yes	Must be positive
unitCost	Decimal	Yes	Must be positive (precision 10, scale 2)
purchaseDate	String	No	ISO datetime; defaults to now

Note: *totalCost is automatically calculated as unitCost * quantity. The status defaults to "ORDERED".*

Success Response (200 OK):

```
{
  "id": 2,
  "product": {
    "id": 1,
    "name": "Wireless Mouse",
    "sku": "WM-001"
  },
  "supplier": {
    "id": 1,
    "name": "TechWholesale Ltd"
  },
  "quantity": 100,
  "unitCost": 15.00,
  "totalCost": 1500.00,
  "purchaseDate": "2026-07-25T10:00:00",
  "status": "ORDERED",
  "createdAt": "2026-07-25T10:00:00",
  "updatedAt": "2026-07-25T10:00:00"
}
```

5.4 Update Purchase Status

PUT /api/purchases/{id}/status?status={status}

Auth required: Yes

Path Parameters:

Param Type Description

id Long Purchase ID

Query Parameters:

Param Type Required

Valid Values

status String Yes ORDERED, SHIPPED, RECEIVED, CANCELLED

Example Request:

```
PUT /api/purchases/1/status?status=RECEIVED
```

Success Response (200 OK):

```
{
  "id": 1,
  "product": {
    "id": 1,
    "name": "Wireless Mouse"
  },
  "quantity": 100,
  "unitCost": 15.00,
  "totalCost": 1500.00,
  "status": "RECEIVED",
  "purchaseDate": "2026-07-01T08:00:00",
  "updatedAt": "2026-07-25T11:00:00"
}
```

5.5 Get Purchases by Date Range

```
GET /api/purchases/date-range?start={start}&end={end}
```

Auth required: Yes

Query Parameters:

Param Type Required

Format

Description

start String Yes YYYY-MM-DD (ISO) Start date

end String Yes YYYY-MM-DD (ISO) End date

Example Request:

```
GET /api/purchases/date-range?start=2026-07-01&end=2026-07-31
```

Success Response (200 OK):

```
[
  {
    "id": 1,
    "product": {
      "id": 1,
      "name": "Wireless Mouse",
      "sku": "WM-001"
    },
    "supplier": {
      "id": 1,
      "name": "TechWholesale Ltd"
    },
    "quantity": 100,
    "unitCost": 15.00,
    "totalCost": 1500.00,
    "purchaseDate": "2026-07-01T08:00:00",
    "status": "RECEIVED"
  }
]
```

6. Dashboard (/api/dashboard)

6.1 Get Dashboard Statistics

GET /api/dashboard/stats

Auth required: Yes

Returns an aggregated overview of the inventory system: total counts, low-stock alerts, and recent activity.

Success Response (200 OK):

```

{
  "totalProducts": 45,
  "totalSuppliers": 8,
  "totalSales": 120,
  "totalPurchases": 35,
  "lowStockCount": 5,
  "recentSales": [
    {
      "id": 118,
      "product": {
        "id": 1,
        "name": "Wireless Mouse"
      },
      "quantity": 3,
      "unitPrice": 29.99,
      "totalPrice": 89.97,
      "customerName": "GadgetWorld",
      "saleDate": "2026-07-25T09:00:00",
      "status": "COMPLETED"
    },
    {
      "id": 117,
      "product": {
        "id": 5,
        "name": "USB Hub"
      },
      "quantity": 10,
      "unitPrice": 19.99,
      "totalPrice": 199.90,
      "customerName": "Tech Supplies Inc",
      "saleDate": "2026-07-24T16:45:00",
      "status": "COMPLETED"
    }
  ],
  "recentPurchases": [
    {
      "id": 30,
      "product": {
        "id": 2,
        "name": "USB-C Cable"
      },
      "supplier": {
        "id": 3,
        "name": "CableCo"
      },
      "quantity": 500,
      "unitCost": 3.50,
      "totalCost": 1750.00,
      "purchaseDate": "2026-07-24T10:00:00",
      "status": "SHIPPED"
    }
  ]
}

```

7. AI Service (/api/ai)

All AI endpoints require JWT authentication. The Spring Boot backend proxies requests to the Python AI microservice running on <http://localhost:5000>.

7.1 Get Demand Forecast

```
GET /ai/forecast/{productId}
```

Auth required: Yes

Path Parameters:

Param	Type	Description
productId	Long	Product ID

Returns an AI-generated demand forecast for the specified product.

Success Response (200 OK):

```
{
  "productId": 1,
  "productName": "Wireless Mouse",
  "forecast": {
    "nextWeek": 25,
    "nextMonth": 100,
    "nextQuarter": 280,
    "trend": "increasing",
    "confidence": 0.87
  },
  "historicalData": {
    "avgDailySales": 3.2,
    "avgWeeklySales": 22.4,
    "peakDay": "Friday",
    "seasonality": "moderate"
  },
  "recommendations": [
    "Maintain current stock level",
    "Consider increasing order quantity by 15%"
  ]
}
```

7.2 Get Reorder Suggestions

GET /ai/reorder-suggestions

Auth required: Yes

Returns AI-generated reorder suggestions for products that are low on stock or predicted to go out of stock soon.

Success Response (200 OK):

```
[
  {
    "productId": 2,
    "productName": "USB-C Cable",
    "sku": "UC-002",
    "currentStock": 12,
    "reorderLevel": 50,
    "suggestedReorderQuantity": 200,
    "urgency": "HIGH",
    "estimatedStockoutDate": "2026-08-01",
    "preferredSupplier": {
      "id": 3,
      "name": "CableCo"
    },
    "estimatedCost": 700.00
  },
  {
    "productId": 5,
    "productName": "USB Hub",
    "sku": "UH-005",
    "currentStock": 8,
    "reorderLevel": 15,
    "suggestedReorderQuantity": 50,
    "urgency": "MEDIUM",
    "estimatedStockoutDate": "2026-08-10",
    "preferredSupplier": {
      "id": 1,
      "name": "TechWholesale Ltd"
    },
    "estimatedCost": 250.00
  }
]
```

7.3 Get Anomalies

GET /ai/anomalies

Auth required: Yes

Detects unusual patterns in inventory movements, sales, or stock levels that may indicate theft, data entry errors, or demand spikes.

Success Response (200 OK):

```
[
  {
    "type": "STOCK_DISCREPANCY",
    "severity": "HIGH",
    "productId": 3,
    "productName": "Wireless Keyboard",
    "message": "Physical count differs from system stock by 15 units",
    "detectedAt": "2026-07-25T08:00:00",
    "details": {
      "systemStock": 45,
      "expectedStock": 45,
      "deviation": 15
    }
  },
  {
    "type": "UNUSUAL_SALE_VOLUME",
    "severity": "MEDIUM",
    "productId": 1,
    "productName": "Wireless Mouse",
    "message": "Sales volume 3x higher than 7-day average",
    "detectedAt": "2026-07-25T10:00:00",
    "details": {
      "todaySales": 30,
      "sevenDayAvg": 10,
      "multiplier": 3.0
    }
  }
]
```

7.4 Get Stock Health Report

GET /ai/stock-health

Auth required: Yes

Returns an overall health assessment of inventory stock levels across all products.

Success Response (200 OK):

```
{
  "overallScore": 78,
  "totalProducts": 45,
  "healthyStock": 35,
  "lowStock": 7,
  "overstocked": 3,
  "categories": {
    "electronics": {
      "score": 82,
      "products": 20,
      "issues": 3
    },
    "accessories": {
      "score": 71,
      "products": 15,
      "issues": 5
    },
    "furniture": {
      "score": 85,
      "products": 10,
      "issues": 1
    }
  },
  "recommendations": [
    "7 products are below reorder level - immediate restocking recommended",
    "3 products are overstocked - consider promotional pricing",
    "Overall inventory turnover rate is healthy at 4.2x per quarter"
  ]
}
```

7.5 Natural Language Query

POST /ai/query

Auth required: Yes

Submit a natural language question about inventory data and receive a structured response.

Request Body:

```
{
  "query": "What are the top 5 selling products this month?"
}
```

Field	Type Required	Description
query	String Yes	Natural language question

Success Response (200 OK):

```
{
  "query": "What are the top 5 selling products this month?",
  "answer": "The top 5 selling products this month are: 1. Wireless Mouse (120 units), 2. USB-C Cable (95 units), 3. Wireless Keyboard (72 units), 4. USB Hub (58 units), 5. Monitor Stand (41 units).",
  "data": [
    {
      "rank": 1,
      "productId": 1,
      "productName": "Wireless Mouse",
      "totalSold": 120
    },
    {
      "rank": 2,
      "productId": 2,
      "productName": "USB-C Cable",
      "totalSold": 95
    },
    {
      "rank": 3,
      "productId": 3,
      "productName": "Wireless Keyboard",
      "totalSold": 72
    },
    {
      "rank": 4,
      "productId": 5,
      "productName": "USB Hub",
      "totalSold": 58
    },
    {
      "rank": 5,
      "productId": 8,
      "productName": "Monitor Stand",
      "totalSold": 41
    }
  ],
  "confidence": 0.92
}
```

Appendix A: Status Codes

Code	Meaning
200	Success
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error

Appendix B: Data Types Reference

Type	Format / Range
Long	64-bit integer
Integer	32-bit integer
Decimal	BigDecimal(10,2) -- up to 8 digits before decimal
String	UTF-8 text
Boolean	true / false
DateTime	ISO 8601 format: YYYY-MM-DDTHH:MM:SS
Date	ISO 8601 format: YYYY-MM-DD

Appendix C: Movement Types

Value	Description
IN	Stock added to inventory (purchase, return)

Value	Description
OUT	Stock removed from inventory (sale, damage)

Appendix D: Purchase Status Flow



Appendix E: Sale Status Values

Value	Description
PENDING	Sale created, not finalized
COMPLETED	Sale finalized and recorded
CANCELLED	Sale was cancelled

Appendix F: User Roles

Role	Description
USER	Default role; basic CRUD operations
MANAGER	Can view reports and manage inventory
ADMIN	Full access to all endpoints and user management